(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION

(B. Tech CSE/IT/MAE/DMAM/ECE-AI/Reappear) (Semester : 4th)

(March, 2024) OFF LINE mode

Subject Code: BIT 204	Subject: Object Oriented Programming
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Explain the concept of dynamic memory allocation in C++.	UNIT-1	
	(b) What are default arguments and how they are used in function call?	UNIT-1	
	(c) What is the difference between base class and derived class? How base class members can be accessed by derived class?	UNIT-2	
	(d) Explain the polymorphism and data hiding features of C++.	UNIT-2	
Q2	(Attempt any Two Parts) UNIT-1		(5,5)
	(a) Write a program using class to implement the basic functions of a calculator.		
	(b) Explain any 2 usage of the keyword static through suitable programs for each. OR Enlist the difference between memory allocated on the stack and memory allocated on the heap in C++.		
	(C) How does a friend function differ from a member function in a class? Write a Program to swap private data members of classes named as ***class1***, ***class2*** using friend function.		
			(5,5)
Q3	(Attempt any Two Parts) UNIT-2		
	(a) Write a program that calculates the area of rectangle or circle and print the area using overloading concept.		
	(b) Write a Program to design a student class representing student roll no. and a test class (derived class of student) representing the scores of the student in two subjects and sports class (derived class of student) representing the score in sports. The sports and test class should be inherited by a result class having the functionality to add the scores and display the result for a student.		
	(c) Write a program to demonstrate the usage of virtual base class.		